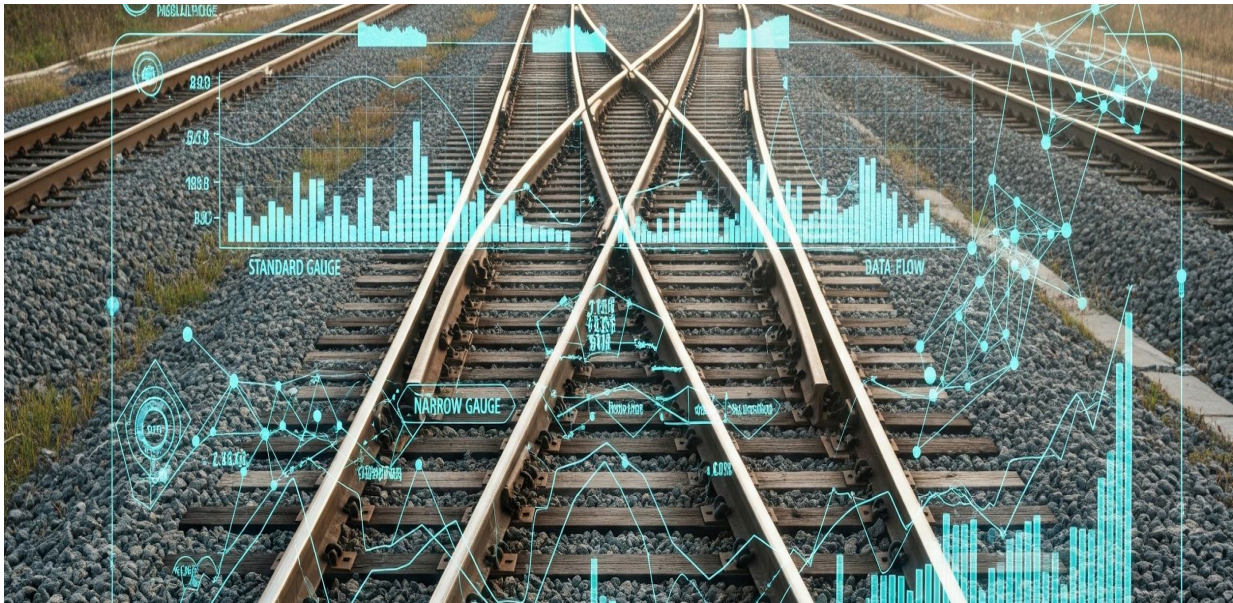




Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges.csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is titled **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.

Modules Used

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (● Easy)

Description:

This scenario is about loading and preparing the railway dataset for analysis. The data is read from a CSV file using Pandas. The first 5 rows and column names are displayed to understand the data. Then the dataset is checked for missing values and replaced with 0 if any are found. Finally, the gauge columns are converted into numeric types for easy calculations.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:

The dataset was loaded and cleaned successfully. There were no missing values, so the data was already in good condition. All required columns were converted into numeric format. This makes the dataset ready for further analysis and visualization. This step is important for getting correct results in the next scenarios.

Scenario 2: Simple Visualization (● Easy)

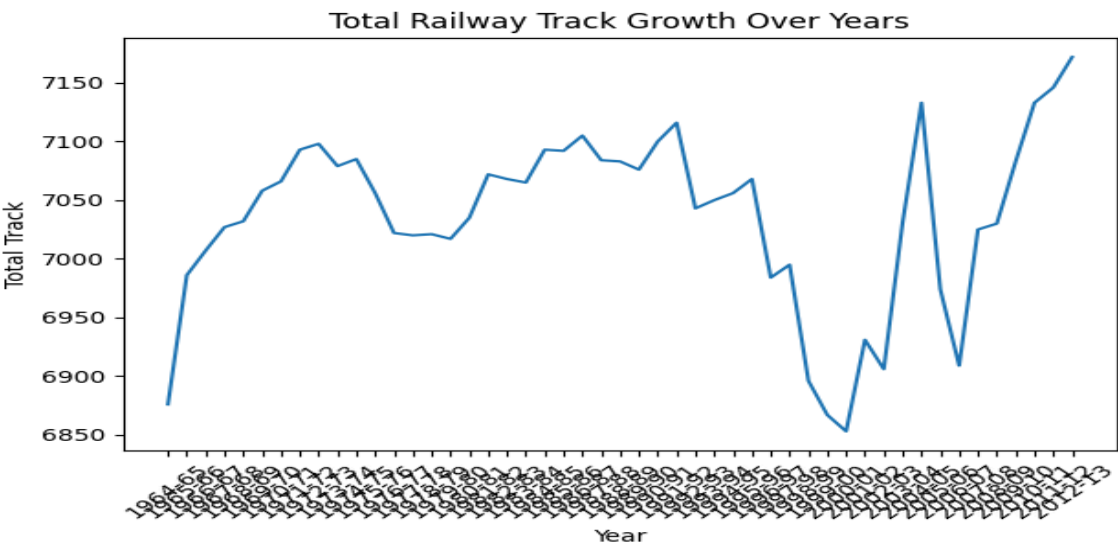
Description:

This scenario focuses on understanding how total railway track length has changed over time. The Year and Total columns are selected from the dataset using Pandas. A line graph is plotted using Matplotlib to show the change in total tracks across years. Titles and labels are added to make the graph clear and easy to read. This helps in identifying the overall trend of railway growth.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph :

The graph is a line chart where the X-axis represents the years and the Y-axis represents the total railway track length. Each point on the graph shows the total track length for a particular year, and the points are connected with a line. The line moves upward over time, showing how the railway network has grown. The graph clearly helps in visualizing changes and trends across different years.



Conclusion:

The line graph shows that the total railway track length is increasing over the years. This indicates steady growth in railway infrastructure. The upward trend suggests continuous development and expansion. Overall, the railway network has improved over time.

Scenario 3: Filtering + Bar Chart (🟡 Medium)

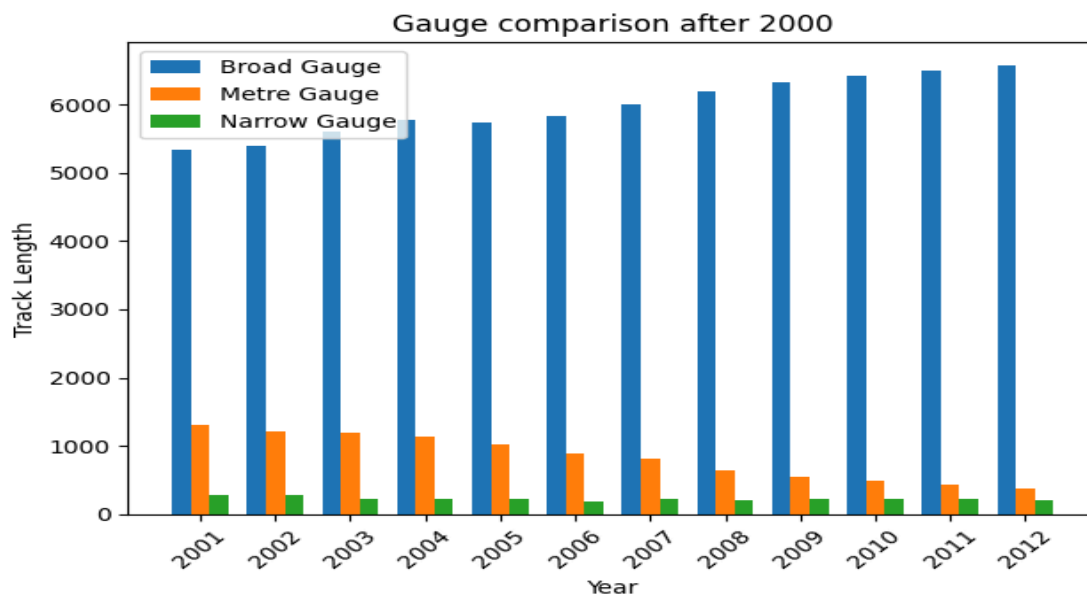
Description:

This scenario focuses on analyzing recent railway expansion after the year 2000. The dataset is filtered to include only years greater than 2000 using Pandas. The Broad Gauge, Metre Gauge, and Narrow Gauge columns are selected for comparison. A grouped bar chart is created using Matplotlib to compare these gauges across different years. Labels and a legend are added to make the chart easy to understand.

Task	Description
1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph :

The graph is a grouped bar chart where the X-axis represents the years after 2000 and the Y-axis represents the track length. For each year, three bars are shown side by side, representing Broad Gauge, Metre Gauge, and Narrow Gauge. This makes it easy to compare the values of each gauge for the same year. The legend helps in identifying each gauge type clearly.



Conclusion:

The bar chart shows that Broad Gauge has higher values compared to Metre and Narrow Gauge in recent years. This indicates that Broad Gauge is the dominant type in modern railway expansion. Metre and Narrow Gauge have lower contributions. Overall, railway development is mainly focused on Broad Gauge.

Scenario 4: Feature Engineering + Pie Chart (🟡 Medium)

Description:

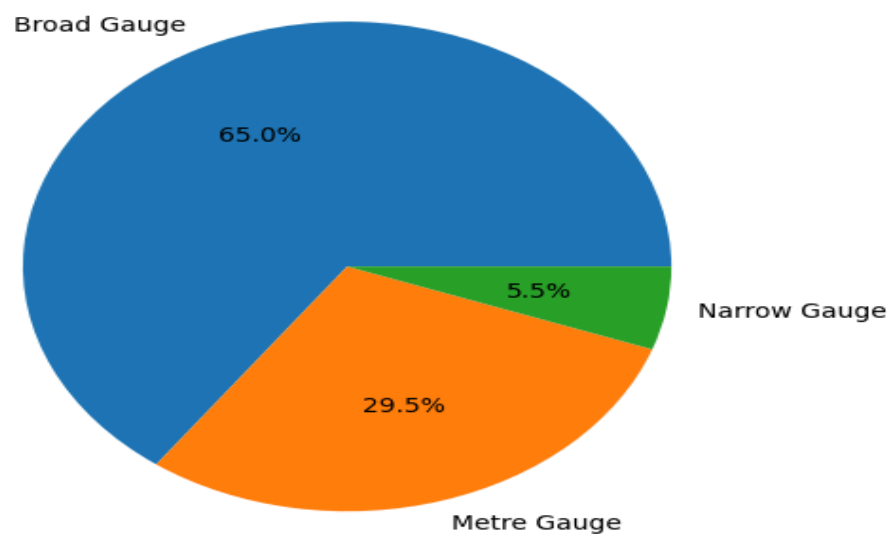
This scenario focuses on understanding how much each railway gauge contributes to the total track length. The total sum of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated using Pandas. A new structure is created to store these totals. A pie chart is then plotted using Matplotlib to show the percentage contribution of each gauge. Percentage labels are added to make the chart easy to understand.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:

The graph is a pie chart that shows the percentage share of each gauge type. Each slice of the pie represents one gauge (Broad, Metre, or Narrow). The size of each slice depends on its total contribution. Percentage values are displayed on the chart, making it easy to compare which gauge contributes more. Larger slices indicate higher contribution.

Percentage contribution of each gauge type



Conclusion:

The pie chart shows that Broad Gauge has the highest contribution among all gauge types. Metre and Narrow Gauge have smaller shares. This indicates that Broad Gauge plays the most important role in the railway network. Overall, the majority of railway tracks belong to Broad Gauge.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

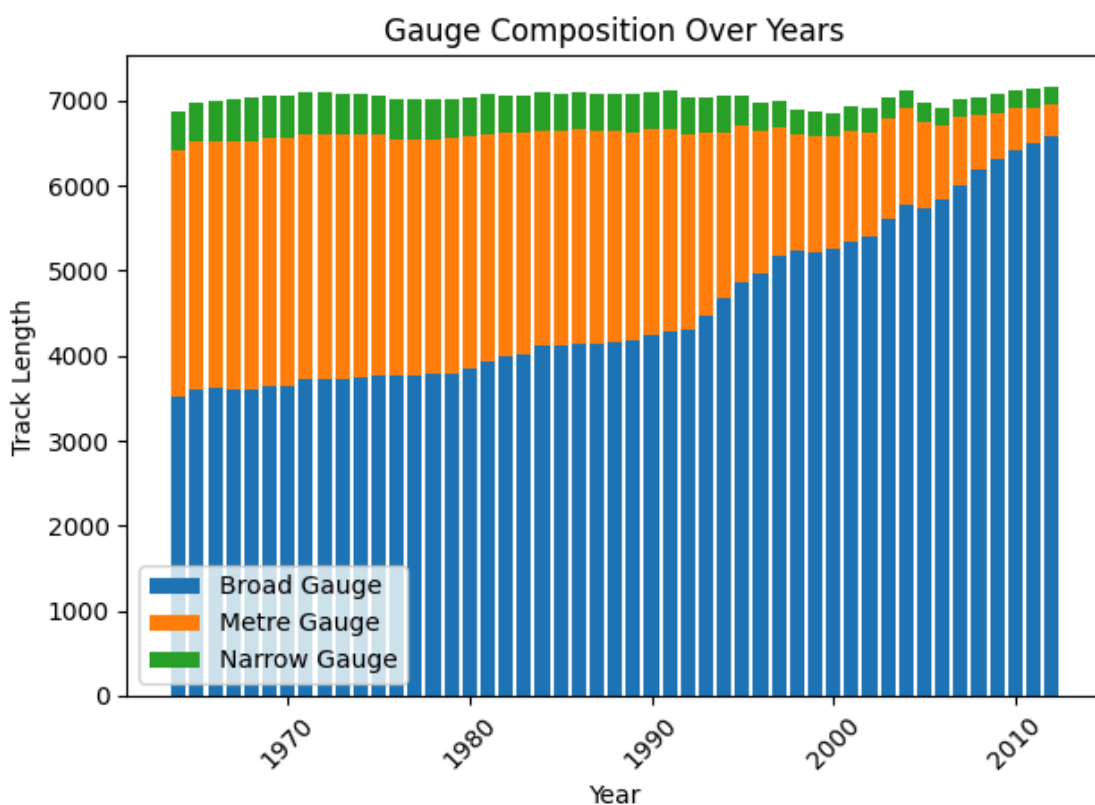
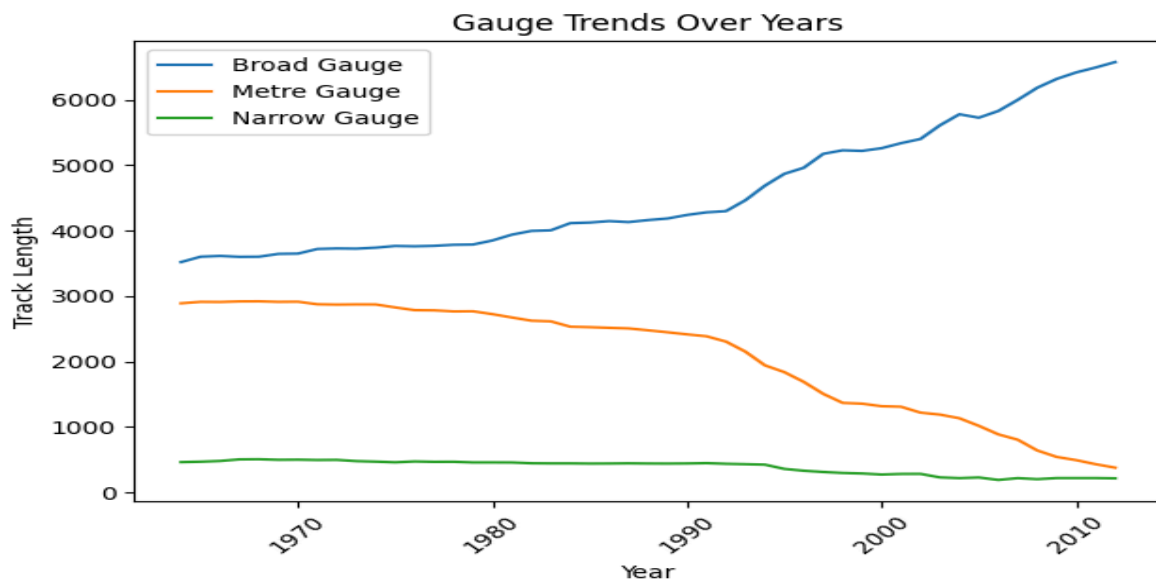
Description:

This scenario focuses on performing a complete analysis of railway trends. New columns are created to calculate the percentage of Broad, Metre, and Narrow Gauge using Pandas. Yearly growth of total railway tracks is calculated using NumPy. A line graph and a stacked bar chart are plotted using Matplotlib to visualize trends and composition. The analysis helps identify years with highest growth and any decline in gauges.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function <code>np.diff</code> to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Graph :

The first is a **line graph** showing the trend of Broad, Metre, and Narrow Gauge over the years. Each line represents one gauge, making it easy to compare their growth patterns. The second is a **stacked bar chart**, where each bar represents a year and is divided into parts showing the contribution of each gauge. This helps in understanding how the total track is composed of different gauges.



Conclusion:

The analysis shows that Broad Gauge has increased steadily over time, while Metre and Narrow Gauge have either decreased or grown slowly. Some years show higher growth in total tracks, indicating expansion periods. The overall trend suggests that the railway system is shifting towards a single dominant gauge, which is Broad Gauge. This indicates modernization and standardization in railway infrastructure.

Final Conclusion:

This project helped in analyzing railway gauge data step by step. The data was first cleaned and prepared for analysis. Different graphs like line chart, bar chart, and pie chart were used to understand trends and comparisons. The analysis showed that railway tracks have increased over time. It also showed that Broad Gauge is the most used and dominant type. Overall, the project gives a clear idea about railway growth and development.

What you have learned

In this project, I learned how to work with data using Pandas. I understood how to clean data, handle missing values, and convert data into proper format. I also learned how to use NumPy for calculations and Matplotlib for creating different types of graphs. This project helped me understand how to analyze data and draw conclusions from it. It improved my practical knowledge of data analysis using Python.